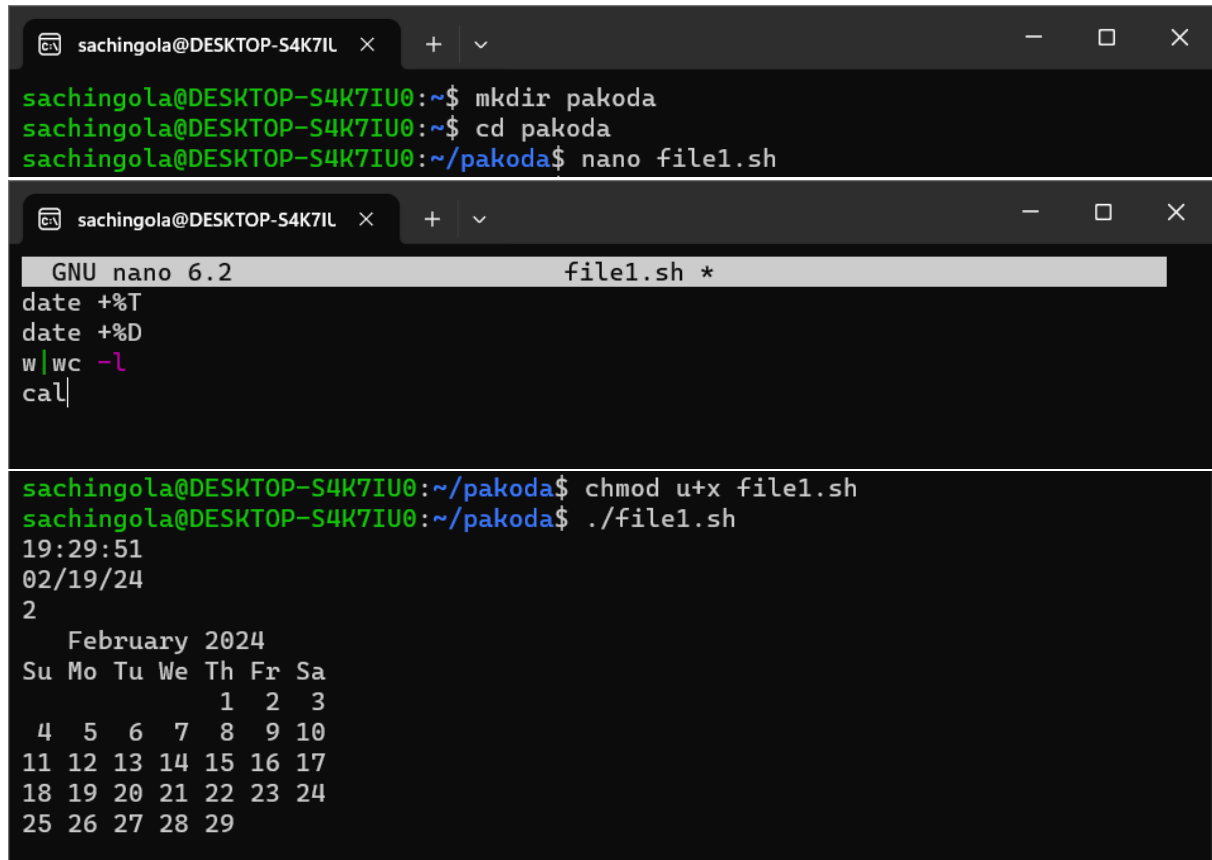


OS-LAB ASSIGNMENT-03

1. Write a shell script to display current date in a particular format, number of users currently login and current month's calendar.

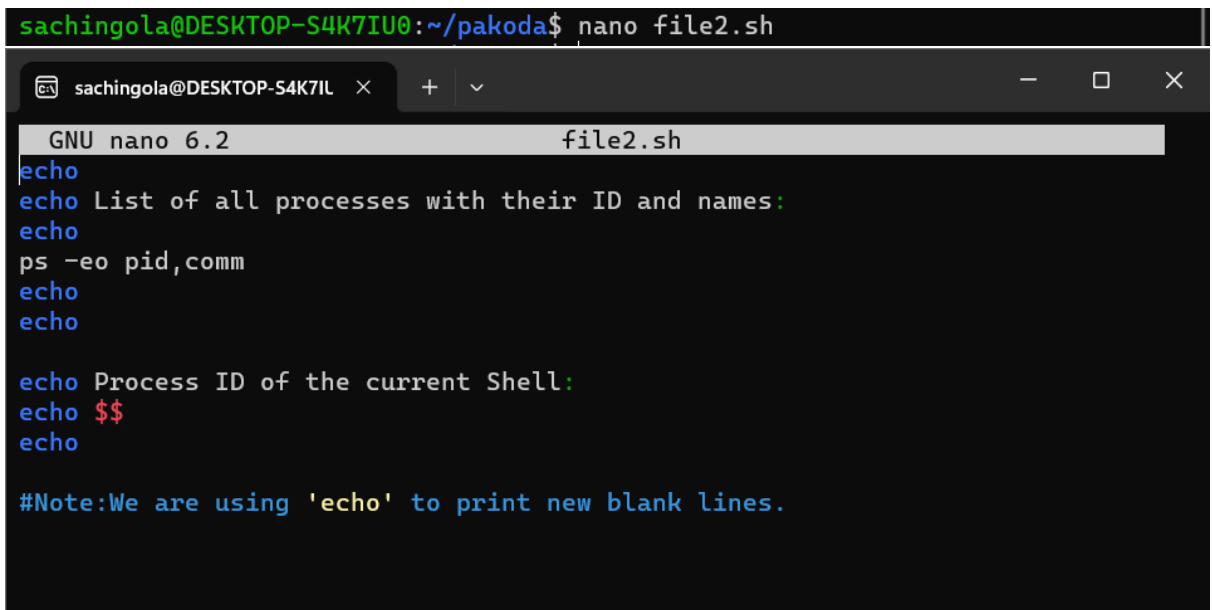


```
sachingola@DESKTOP-S4K7IIL x + v
sachingola@DESKTOP-S4K7IU0:~$ mkdir pakoda
sachingola@DESKTOP-S4K7IU0:~$ cd pakoda
sachingola@DESKTOP-S4K7IU0:~/pakoda$ nano file1.sh

GNU nano 6.2 file1.sh *
date +%T
date +%D
w|wc -l
cal|

sachingola@DESKTOP-S4K7IU0:~/pakoda$ chmod u+x file1.sh
sachingola@DESKTOP-S4K7IU0:~/pakoda$ ./file1.sh
19:29:51
02/19/24
2
February 2024
Su Mo Tu We Th Fr Sa
      1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29
```

2. Write a shell script to display the process name and its process id.



```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ nano file2.sh

GNU nano 6.2 file2.sh
echo
echo List of all processes with their ID and names:
echo
ps -eo pid,comm
echo
echo

echo Process ID of the current Shell:
echo $$
echo

#Note:We are using 'echo' to print new blank lines.
```

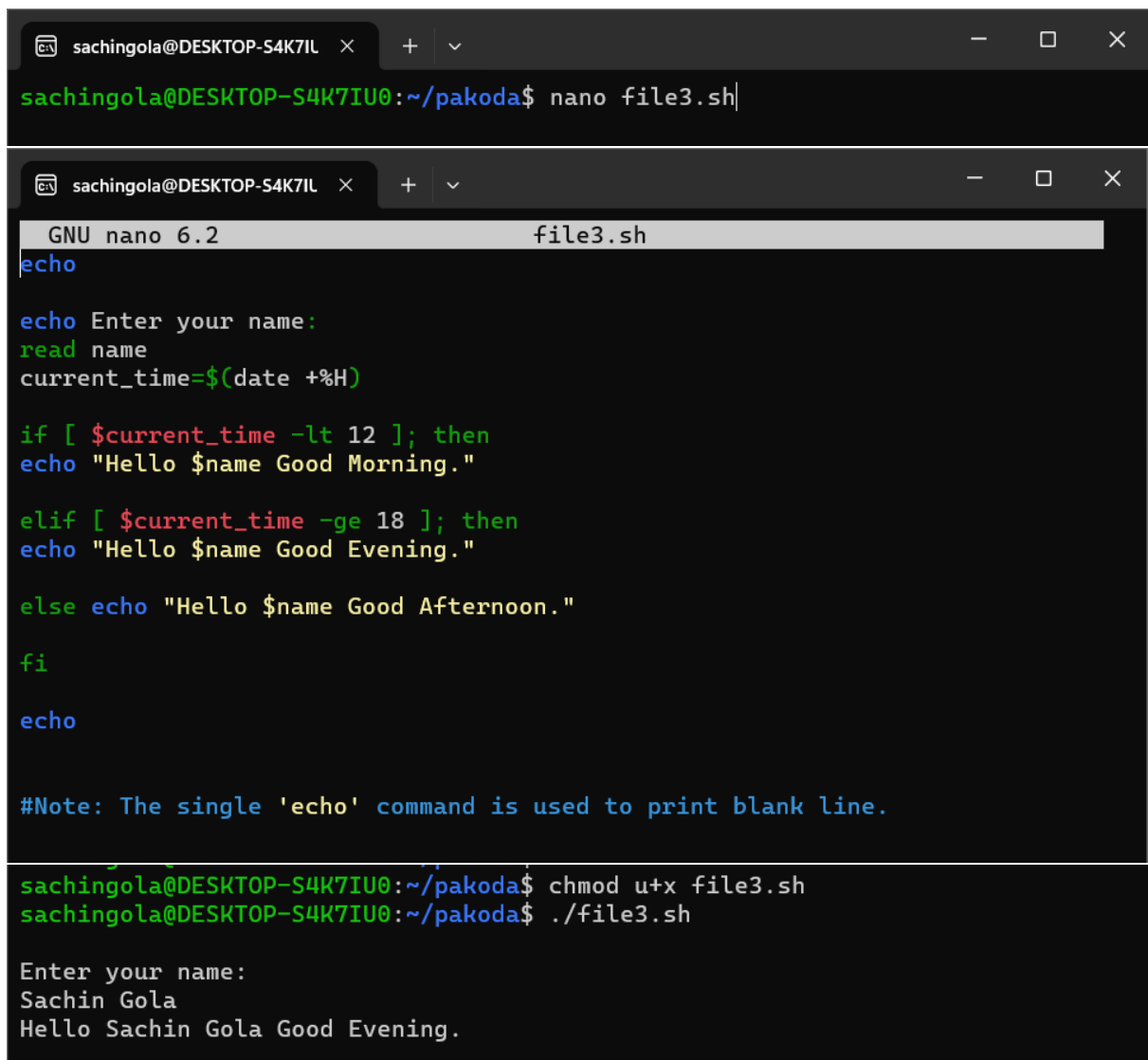
```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ chmod u+x file2.sh
sachingola@DESKTOP-S4K7IU0:~/pakoda$ ./file2.sh

List of all processes with their ID and names:

PID COMMAND
  1 init
 13 init
 14 init
 15 bash
 91 bash
 92 ps

Process ID of the current Shell:
91
```

3. Write a shell script to take name as a input and display a greeting message to the user by checking system clock.



```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ nano file3.sh
```

```
GNU nano 6.2 file3.sh
echo

echo Enter your name:
read name
current_time=$(date +%H)

if [ $current_time -lt 12 ]; then
echo "Hello $name Good Morning."

elif [ $current_time -ge 18 ]; then
echo "Hello $name Good Evening."

else echo "Hello $name Good Afternoon."
fi

echo

#Note: The single 'echo' command is used to print blank line.
```

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ chmod u+x file3.sh
sachingola@DESKTOP-S4K7IU0:~/pakoda$ ./file3.sh

Enter your name:
Sachin Gola
Hello Sachin Gola Good Evening.
```

4. Write a shell script to merge the content of 2 files into one file.

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ cat > first.txt
If nothing goes right , go left.
^C
sachingola@DESKTOP-S4K7IU0:~/pakoda$ cat > second.txt
If you are in hell, good luck and continue to be in.
^C

sachingola@DESKTOP-S4K7IU0:~/pakoda$ touch merged_file.txt
sachingola@DESKTOP-S4K7IU0:~/pakoda$ nano file4.sh

GNU nano 6.2 file4.sh
# Receiving names of files( must exist) from user in three variables :
echo
read file1
read file2
read merged_file

# Merging the content of the files into one:

cat "$file1" "$file2" > "$merged_file"
echo

echo "The contents of given files have been merged. The content of merged file is merged file."
echo

cat "$merged_file"
echo

sachingola@DESKTOP-S4K7IU0:~/pakoda$ chmod u+x file4.sh
sachingola@DESKTOP-S4K7IU0:~/pakoda$ ./file4.sh

first.txt
second.txt
merged_file.txt

The contents of given files have been merged. The content of merged file is merged file.

If nothing goes right , go left.
If you are in hell, good luck and continue to be in.
```

5. Write a shell script to create a tsv file containing name, roll no. and age of 10 students. Then use that tsv file to display only the names of the students in alphabetical order.

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ nano file5.sh
sachingola@DESKTOP-S4K7IU0:~/pakoda$ touch new_file.tsv

GNU nano 6.2 file5.sh *
#Given file must exist
filename="new_file.tsv"

echo -e "Name\tRoll_No\tAge" > "$filename"
echo

#Using For loop to get data from user:

for ((i=1; i<=10; i++)); do
read -p "Name: " data1
read -p "Roll number: " data2
read -p "Age : " data3

# Appending data in file after each iteration:
echo -e "$data1\t$data2\t$data3" >> "$filename"
echo
done

#printing names in alphabetical order:

echo "Names of students in alphabetical order:"

cut -f 1 "$filename" | sort
```

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ ./file5.sh
```

```
Name: Aman  
Roll number: 1  
Age :19
```

```
Name: Bhawna  
Roll number: 2  
Age :20
```

```
Name: Chetan  
Roll number: 3  
Age :18
```

```
Name: Farhan  
Roll number: 4  
Age :21
```

```
Name: Gauri  
Roll number: 6  
Age :19
```

```
Name: Himani  
Roll number: 7  
Age :19
```

```
Name: Jagdish  
Roll number: 8  
Age :20
```

```
Name: Mayank  
Roll number: 9  
Age :18
```

```
Name: Payal  
Roll number: 10  
Age :20
```

```
Name: Raunak  
Roll number: 11  
Age :19
```

```
Names of students in alphabetical order:
```

```
Aman  
Bhawna  
Chetan  
Farhan  
Gauri  
Himani  
Jagdish  
Mayank  
Name  
Payal  
Raunak
```

6. You are given a file of tab-delimited weather data (TSV). There is no header column in this data file. The first five columns of this data are: (a) the name of the city (b) the average monthly temperature in Jan (in Fahrenheit). (c) The average monthly temperature in April (in Fahrenheit). (d) The average monthly temperature in July (in Fahrenheit). (e) the average monthly temperature in October (in Fahrenheit). You need to sort this file on the basis of average monthly temperature in April.

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ touch sixth_file.tsv  
sachingola@DESKTOP-S4K7IU0:~/pakoda$ nano file6.sh
```

GNU nano 6.2

file6.sh

#Given file must exist

filename="sixth_file.tsv"

echo

#Using For loop to get data from user:

for ((i=1; i<=5; i++)); do

read -p "Name of the city: " data1

read -p "Avg. temp. of January(in F): " data2

read -p "Avg. temp. of April(in F): " data3

read -p "Avg. temp. of July(in F): " data4

read -p "Avg. temp. of October(in F): " data5

Appending data in file after each iteration:

echo -e "\$data1\t\$data2\t\$data3\t\$data4\t\$data5" >> "\$filename"

echo

done

echo "Sorting file on the basis of average temperature in April: "

echo

sort -k3,3 -n "\$filename"

sachingola@DESKTOP-S4K7IU0:~/pakoda\$ chmod u+x file6.sh

```
sachingola@DESKTOP-S4K7IU0:~/pakoda$ ./file6.sh
```

```
Name of the city: Agra
Avg. temp. of January(in F): 40
Avg. temp. of April(in F): 85
Avg. temp. of July(in F): 96
Avg. temp. of October(in F): 48
```

```
Name of the city: Delhi
Avg. temp. of January(in F): 35
Avg. temp. of April(in F): 90
Avg. temp. of July(in F): 100
Avg. temp. of October(in F): 58
```

```
Name of the city: Sydney
Avg. temp. of January(in F): 78
Avg. temp. of April(in F): 85
Avg. temp. of July(in F): 46
Avg. temp. of October(in F): 87
```

```
Name of the city: Dubai
Avg. temp. of January(in F): 78
Avg. temp. of April(in F): 96
Avg. temp. of July(in F): 89
Avg. temp. of October(in F): 85
```

```
Name of the city: Ambala
Avg. temp. of January(in F): 30
Avg. temp. of April(in F): 50
Avg. temp. of July(in F): 74
Avg. temp. of October(in F): 64
```

```
Sorting file on the basis of average temperature in April:
```

Ambala	30	50	74	64
Agra	40	85	96	48
Sydney	78	85	46	87
Delhi	35	90	100	58
Dubai	78	96	89	85